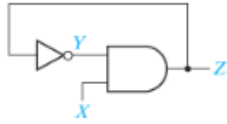
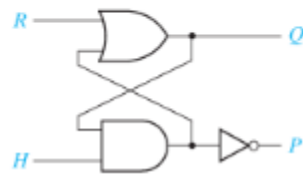


11.1 – 11.3, 11.7, 11.9, 11.10, 11.16, 11.21, 11.22, 11.26, 11.27

11.1 Assume that the inverter in the given circuit has a propagation delay of 5 ns and the AND gate has a propagation delay of 10 ns. Draw a timing diagram for the circuit showing X , Y , and Z . Assume that X is initially 0, Y is initially 1, after 10 ns X becomes 1 for 80 ns, and then X is 0 again.



11.2 A latch can be constructed from an **OR** gate, an **AND** gate, and an inverter connected as follows:



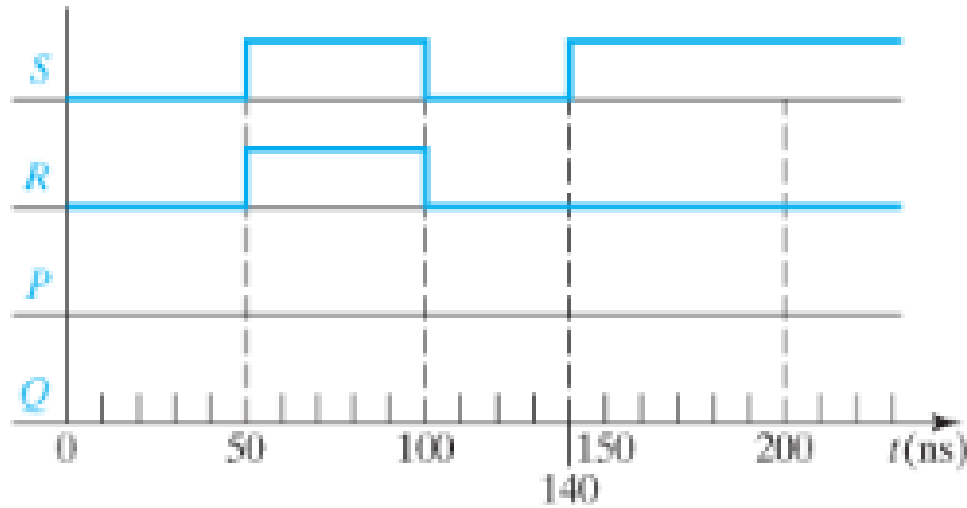
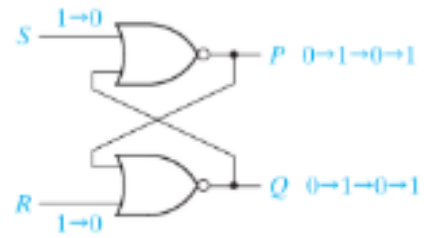
a. What restriction must be placed on R and H so that P will always equal Q' (under steady-state conditions)?

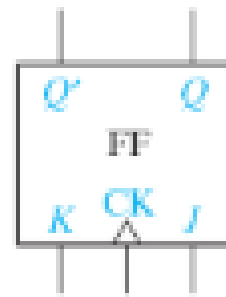
b. Construct a next-state table and derive the characteristic (next-state) equation for the latch.

c. Complete the following timing diagram for the latch.



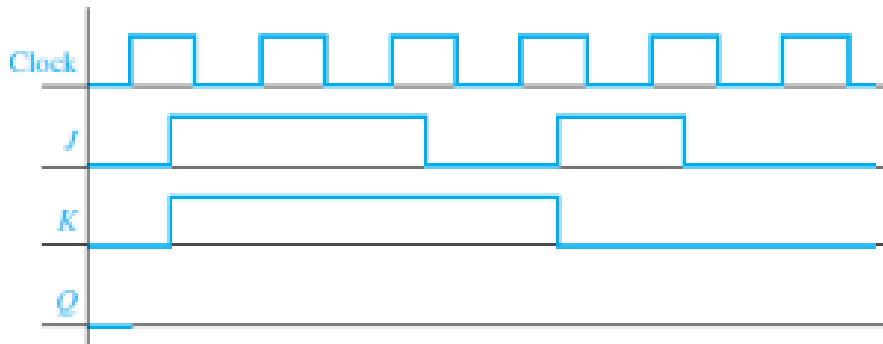
11.3 This problem illustrates the improper operation that can occur if both inputs to an S-R latch are 1 and are then changed back to 0. For Figure 11-6, complete the following timing chart, assuming that each gate has a propagation delay of exactly 10 ns. Assume that initially $P = 1$ and $Q = 0$. Note that when $t = 100$ ns, S and R are both changed to 0. Then, 10 ns later, both P and Q will change to 1. Because these 1's are fed back to the gate inputs, what will happen after another 10 ns?





11.7 Complete the following timing diagram for the flip-flop of [Figure 11-24\(a\)](#).

(a) J-K flip-flop

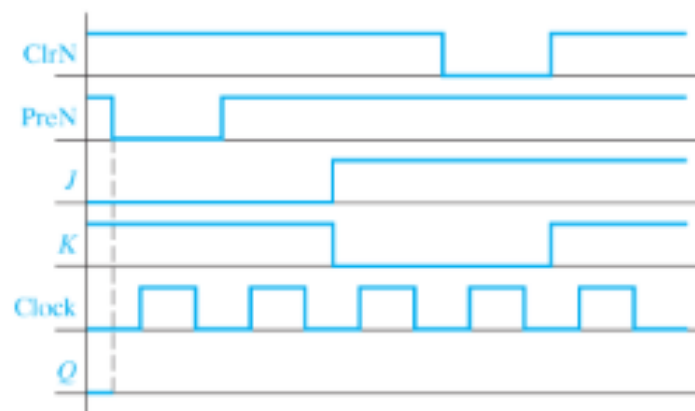


J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

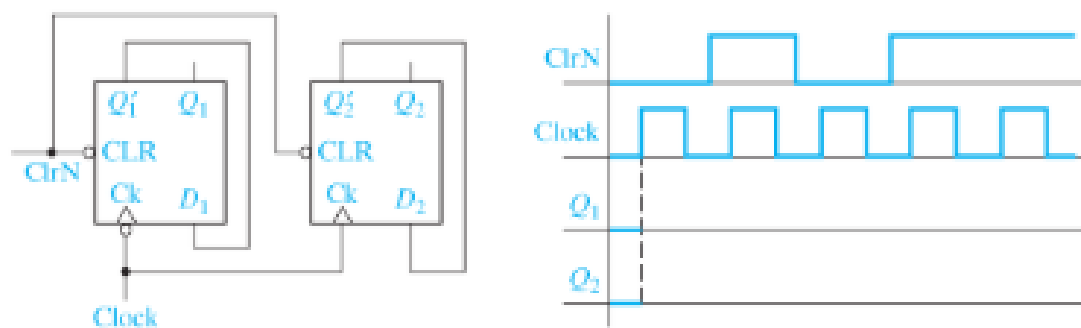
$$Q^+ = JQ' + K'Q$$

(b) Truth table and characteristic equation

- 11.9 a. Complete the following timing diagram for a J-K flip-flop with a falling-edge trigger and asynchronous ClrN and PreN inputs.



- b. Complete the timing diagram for the following circuit. Note that the Ck inputs on the two flip-flops are different.

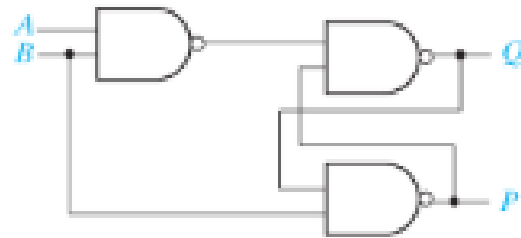


11.10 Convert by adding external gates:

a. a D flip-flop to a J-K flip-flop.

b. a T flip-flop to a D flip-flop.

c. a T flip-flop to a D flip-flop with clock enable.



11.16 Analyze the latch circuit shown.

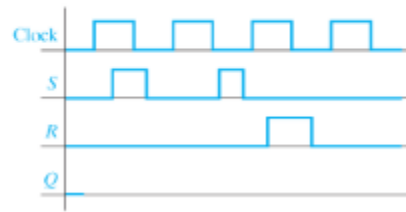
- a. Derive the next-state equation for this circuit using Q as the state variable and P as an output.

- b. Construct the state table and output table for the circuit. Circle the stable states of the circuit.

- c. Are there any restrictions on the allowable input combinations on A and B ? Explain your answer.

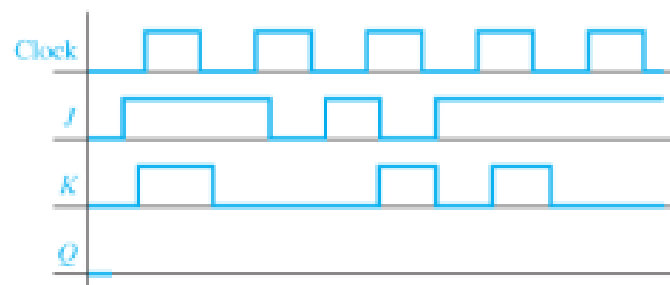
- d. Is the output P usable as the complement of Q ? Verify your answer.

11.21 Fill in the timing diagram for a falling-edge-triggered S-R flip-flop. Assume Q begins at 0 .

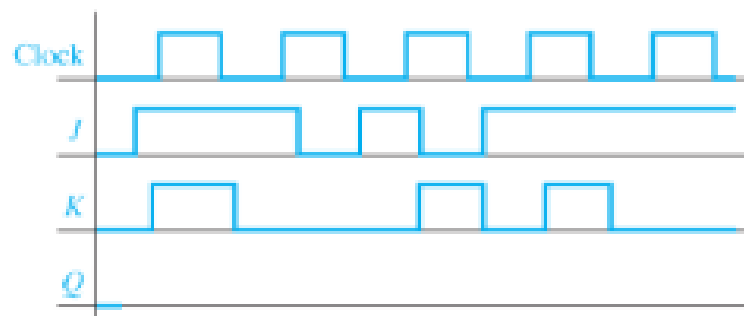


11.22 Fill in the timing diagram for a falling-edge-triggered J-K flip-flop.

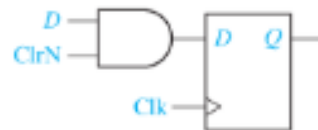
a. Assume Q begins at 0 .



b. Assume Q begins at 1 , but Clock, J , and K are the same.



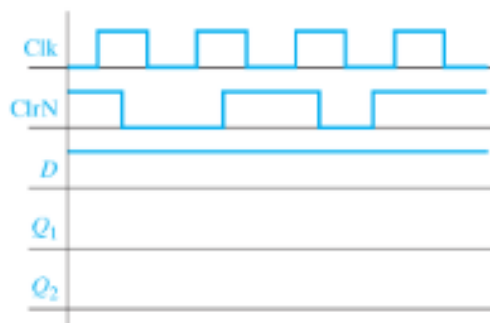
11.26 The ClrN and PreN inputs introduced in [Section 11.8](#) are called asynchronous because they operate independently of the clock (i.e., they are not synchronized with the clock). We can also make flip-flops with synchronous clears or preset inputs. A D -flip-flop with an active-low synchronous ClrN input may be constructed from a regular D flip-flop as follows.



▼ Details

A circuit diagram consists of an AND gate a flip flop. The AND gate receives two inputs, D and Cl r N . The F F receives two inputs, D (the output from the AND gate) and C l k (with a triangle) and provides an output, Q .

Fill in the timing diagram. For Q_1 , assume a synchronous ClrN as above, and for Q_2 , assume an asynchronous ClrN as in [Section 11.8](#). Assume $Q_1 = Q_2 = 0$ at the beginning.



11.27 a. Construct a D flip-flop using an inverter and an S-R flip-flop.

b. If the propagation delay and setup time of the S-R flip-flop in (a) are 2.5 ns and 1.5 ns, respectively, and if the inverter has a propagation delay of 1 ns, what are the propagation delay and setup time of the D flip-flop of part (a)?